

Project 3 Documentation

Project Title:

Multilayer Switch Network with Inter-VLAN Routing and Router-based DHCP

Objective

The objective of this project was to design and implement a small enterprise network using Cisco Packet Tracer. The network separates different departments into VLANs while allowing communication between them through a Multilayer Switch. A router is used as the DHCP server to automatically assign IP addresses to end devices.

Network Requirements

The network should satisfy the following requirements:

- Four separate departments:
 - HR
 - Sales
 - IT
 - Management
- Each department should belong to its own VLAN.
- Inter-VLAN communication should be enabled.
- A Router should act as the DHCP server.
- DHCP requests should be forwarded by the Multilayer Switch using DHCP Relay (IP Helper Address).
- Appropriate subnetting and IP addressing should be implemented.
- All devices should be connected using appropriate cables.
- End devices should automatically receive IP addresses through DHCP.
- All departments should be able to communicate with each other.

Devices Used

Device	Quantity
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Router	1
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Multilayer Switch	1
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Layer 2 Switch	1
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PCs	8
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Printer	1
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Server	1
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VLAN Configuration

VLAN	Department
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VLAN 10	HR
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VLAN 20	Sales
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VLAN 30	IT
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VLAN 40	Management
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IP Addressing Scheme

VLAN 10 (HR)

Network:

192.168.60.0/29

Gateway:

192.168.60.6

VLAN 20 (Sales)

Network:

192.168.60.8/29

Gateway:

192.168.60.14

VLAN 30 (IT)

Network:

192.168.60.16/29

Gateway:

192.168.60.22

VLAN 40 (Management)

Network:

192.168.60.24/29

Gateway:

192.168.60.30

Router to Multilayer Switch Link

Network:

192.168.60.32/30

Router:

192.168.60.33

Multilayer Switch:

192.168.60.34

Network Design

The network consists of:

- A Layer 2 Switch for HR and Sales departments.
- A Multilayer Switch for:
 - IT Department
 - Management Department
 - Inter-VLAN Routing
 - DHCP Relay
- A Router configured as the DHCP Server.
- Trunk link between both switches.
- Routed link between the Router and Multilayer Switch.

Configuration Summary

The following technologies were implemented:

VLAN Creation

- VLAN 10
- VLAN 20
- VLAN 30
- VLAN 40

Access Ports

Configured for end devices.

Trunk Ports

Configured between both switches.

Inter-VLAN Routing

Configured using Switch Virtual Interfaces (SVIs).

DHCP Relay

Configured using:

ip helper-address

on all VLAN interfaces.

DHCP Server

Configured on the Router with separate DHCP pools for all VLANs.

Static Routing

Configured between the Router and the Multilayer Switch.

Connectivity Testing

The following tests were performed successfully:

- ✓ PCs obtained IP addresses automatically.
- ✓ End devices received correct default gateways.

- ✓ Devices successfully pinged their respective gateways.
- ✓ Inter-VLAN communication was successful.
- ✓ Router and Multilayer Switch connectivity was verified.
- ✓ DHCP address allocation was successfully tested.

Skills Learned

This project helped develop practical experience with:

- VLAN configuration
- Access and trunk port configuration
- Multilayer switching
- Switch Virtual Interfaces (SVIs)
- Inter-VLAN routing
- DHCP server configuration
- DHCP relay using IP Helper Address
- Static routing
- IPv4 subnetting
- Cisco Packet Tracer troubleshooting
- Enterprise network design

Result

The network was successfully implemented according to the project requirements. All VLANs were operational, inter-VLAN communication was achieved, DHCP services were provided by the Router, and end devices successfully obtained IP addresses automatically. The network functioned as intended with proper routing and connectivity between all departments

Conclusion

This project demonstrates the implementation of a small enterprise network using VLAN segmentation, Multilayer Switching, Inter-VLAN Routing, DHCP Relay, and Router-based DHCP services. The project provided practical experience in configuring and

troubleshooting multiple Cisco networking technologies commonly used in enterprise environments.